Progressive Education Society's
Modern College of Engineering, Shivajinagar, Pune-05.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)
Department of MCA

Course Detail Document

Academic Year: 2025– 2026

Course: Web Programming

Term: I

Class: S.Y.MCA Division: A & B

Name of the Course Instructor Name of the HOD

Mrs. Swati D. Ghule, Mr. Y. L. Puranik, Dr.Prakash Kene

Vision of the Institute

- Creation of a collaborative academic environment to foster professional excellence and ethical values.

Mission of the Institute

- To develop outstanding professionals with high ethical standards capable of creating and managing global enterprises.
- To foster innovation and research by providing a stimulating learning environment.
- To ensure equitable development of students of all ability levels and backgrounds.
- To be responsive to changes in technology, socio-economic and environmental conditions.
- To foster and maintain mutually beneficial partnerships with alumni and industry.

Objectives of the Institute

- To develop infrastructure appropriate for delivering quality education
- To develop the overall personality of students who will be innovators and future leaders capable of prospering in their work environment.
- To inculcate ethical standards and make students aware of their social responsibilities.
- Promote close interaction among industry, faculty and students to enrich the learning process and enhance career opportunities.
- Encourage faculty in continuous professional growth through quality enhancement programs and research and development activities.
- Foster a healthy work environment which allows for freedom of expression and protection of the rights of all stakeholders through open channels of communication

Vision of the Department

"To develop Competent Technocrats in the field of Computer applications imbued with human values"

Mission of the Department

- To impart knowledge in the field of Computer applications with a focus on developing the required competencies.
- To improve technical skill of the students through practical and hands-on experience.
- To enhance the quality of the students by collaboration with Alumni and Industry.
- To make students socially responsible citizens.

Program Educational Objectives (PEOs)

PEO 1: Graduates will possess the broad knowledge of computer applications for successful careers in industry.

PEO 2: Graduates will exhibit professionalism, ethical attitude, communication skills, team work in their profession and adapt to current trends by engaging in lifelong learning.

PEO 3: Graduates will contribute as responsible citizens with a commitment to the sustainable development of society.

Program Outcomes (POs)

At the end of both semesters students' will be able to–

PO1: Apply knowledge of mathematics, programming logic and coding fundamentals for solution architecture and problem solving.

PO2: Identify, review, formulate and analyze problems for primarily focusing on customer requirements using critical thinking frameworks.

PO3: Design, develop and investigate problems with as an innovative approach for solutions incorporating ESG/SDG goals.

PO4: Select, adapt and apply modern computational tools such as development of algorithms with an understanding of the limitations including human biases.

PO5: Function and communicate effectively as an individual or a team leader in diverse and multidisciplinary groups. Use methodologies such as agile.

PO6: Use the principles of project management such as scheduling, work breakdown structure and be conversant with the principles of Finance for profitable project management.

PO7: Commit to professional ethics in managing software projects with financial aspects. Learn to use new technologies for cyber security and insulate customers from malware

PO8: Change management skills and the ability to learn, keep up with contemporary technologies and ways of working.

Course Structure

Sr. No.	Course Type	Course Code	Course Title	Teaching Scheme (Hours / Week)		Examination / Assessment Scheme and Marks					Credit Scheme		
						Theory		Practical		Total	L	P	Total
				L	P	CIE	SEE	TW	PR				
1.	MJ		Software Testing And Quality Assurance	3	-	40	60	-	-	**100**	3	-	3
2.	MJ		Data Science	3	-	40	60	-	-	**100**	3	-	3
3.	MJ		Web Programming	3	-	40	60	-	-	**100**	3	-	3
4.	PE		Elective –III	3	-	40	60	-	-	**100**	3	-	3
5.	MJ		Data Science Laboratory	-	4	-	-	25	25	**50**	-	2	2
6.	MJ		Web Programming Laboratory	-	4	-	-	25	25	**50**	-	2	2
7.	MJ		Laboratory Practice-III	-	2	-	-	25	-	**25**	-	1	1
8.	RP		Research Project	-	4	-	-	50	50	**100**	-	4	4
9.	SE		Skill Development -1 (Foreign Language)	-	2	-	-	25	-	**25**	-	1	1
Total Hours/week, Marks, Credits				12	16	160	240	150	100	**650**	**12**	**10**	**22**
TOTAL				28 Hrs/week		400 Marks (TH)		250 Marks (PR)					

List of Elective-III Courses

1	PE	MCA10604 (A) - DL	Deep Learning
2	PE	MCA10604 (B) - CC	Cloud Computing
3	PE	MCA10604 - IR	Information Retrieval

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MCA Department

COURSE DETAILS DOCUMENT

Academic Year: 2025-26 (TERM– I)

Class: SY **Div:** A/B
Course Name: Web Programming
Course Code: MCA01603 **Course No.**:C603 **Teaching Scheme Marking Scheme**
Theory –3 Hrs/week Theory Marks: 100 SEE – 40 SEE –60

Prerequisites of the Course

1. Computer Network
2. Database Management System.
3. Basics of Web.

Course Objectives

- To learn the fundamentals of web essentials and markup languages.
- To use the Client side technologies in web development.
- To use the Server side technologies in web development.
- To understand the web services and frameworks.

Course Outcomes

At the end of the Course, Students will be able to

CO603.1: Explain the concept of static web pages using HTML & CSS**.**

CO603.2: Analyze client side technology: JavaScript in detail.

CO603.3: Illustrate the Conceptual Framework of Bootstrap and WordPress.

CO603.4: Apply recent web technologies: AngularJS and jQuery**.**

CO603.5: Utilize the server side technologies for web development.

CO603.6: Demonstrate use of web Services**.**

CO No.	Year of study 2025-26	Mapping to POs		
		Substantial	Moderate	Low
CO603.1	**Explain** the concept of static web pages using HTML & CSS.	PO1, PO3	PO2,PO4, PO5	PO8
CO603.2	**Analyze** client side technology: JavaScript in detail.	PO2, PO4	PO1, PO3, PO5, PO8	-
CO603.3	**Illustrate** the Conceptual Framework of Bootstrap and WordPress.	-	PO1,PO2, PO3, PO4, PO8	PO5
CO603.4	**Apply** recent web technologies: AngularJS and jQuery.	PO1,PO3	PO2, PO4,PO5, PO8	-
CO603.5	**Utilize** the server side technologies for web development..	-	PO1, PO2, PO3, PO4, PO5	PO8
CO603.6	**Demonstrate** use of web Services.	-	PO1, PO2, PO3, PO4, PO8	PO5

*Mapping details based on CO-PO mapping.

CO to PO mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8
	Foundation Knowledge	Problem analysis	Development of solutions	Modern tool usage	Individual and team work	Project Management and finance	Ethics	Life-long learning
CO603.1	3	2	3	2	2	-	-	1
CO603.2	2	3	2	3	2	-	-	2
CO603.3	2	2	2	2	1	-	-	2
CO603.4	3	2	3	2	2	-	-	2
CO603.5	2	2	2	2	2	-	-	1
CO603.6	2	2	2	2	1	-	-	2
CO603	**2.33**	**2.16**	**2.33**	**2.16**	**1.66**	**-**	**-**	**1.6**

Note: In the table above, mapping strength is defined as:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High) '-': It there is no correlation

Justification of CO to PO Mapping

CO	PO	Strength	Justification
C603.1	PO1	3	Substantially mapped as the students will be able to understand the basic concepts of HTML and CSS in Web Programming which helps to frame real-world scenarios.
	PO2	2	Moderately mapped as the students will be able to contributes to the early stages of identifying and addressing customer needs through static web design..
	PO3	3	Substantially mapped as HTML & CSS skills empower students to design and communicate innovative ideas related to ESG/SDG.
	PO4	2	Moderately mapped as the students understand tool limitations in a practical context..
	PO5	2	Moderately mapped as it provides opportunities for collaborative, team-based learning.
C603.2	PO1	3	Substantially mapped as it encourages students to think about solution structure and modularity.
	PO2	3	Substantially mapped as it enables students to identify, analyze, and solve functional challenges on the front end by applying logical reasoning and customer-focused design.
	PO3	2	Moderately mapped as the students are encouraged to develop applications around ESG/SDG themes, their JS skills enable them to execute those ideas effectively.
	PO4	3	Substantially mapped as Students not only develop algorithms but also learn to evaluate the social impact and design inclusively, addressing human biases.
	PO5	2	Moderately mapped as the JavaScript projects naturally involve teamwork and communication in collaborative settings.
	PO8	2	Moderately mapped as the student's ability to adapt to technological change and fosters the mindset necessary to continuously learn and evolve.
C603.3	PO1	3	Substantially mapped as it actively engages students in

			applying programming logic, coding fundamentals, and structured thinking to build and troubleshoot web solutions.
	PO2	2	Moderately mapped as the Students begin to understand how to match technical capabilities with functional requirements.
	PO3	2	Moderately mapped as enables students to design future projects that align with these values.
	PO4	2	Moderately mapped as the students begin to select and apply computational tools with awareness of their technical and ethical constraints.
	PO5	1	Slightly mapped as the students often requires to work collaboratively, communicate effectively, and engage in agile-inspired development cycles.
	PO8	2	Moderately mapped as students need to engage with current technologies, develop adaptive skills, and cultivate a habit of continuous learning—core elements of change management in a technical context.
C603.4	PO1	3	Substantially mapped as the students requires to practically implement and integrate modern front-end web development frameworks to create dynamic, interactive, and responsive web applications.
	PO2	2	Moderately mapped as the students must often analyze user needs and translate those into technical implementations. This requires **identifying and interpreting customer requirements**.
	PO3	3	Substantially mapped as it enables students to **use modern web development tools to design and develop solutions** that are not only technically sound but can also be **aligned with larger ESG/SDG themes**, promoting innovation with purpose.
	PO4	2	Moderately mapped as the Students learn to choose appropriate technologies based on project requirements, demonstrating the ability to select and adapt modern computational tools.
	PO5	2	Moderately mapped as the Students learn to **coordinate tasks, share codebases, and integrate components**, fostering teamwork.

	PO8	2	Moderately mapped as the Students must **learn new syntax, APIs, and development practices**, fostering the ability to adapt to technology changes.
C603.5	PO1	3	Substantially mapped as the students are inherently required to use mathematics, programming logic, and coding fundamentals to architect and implement reliable, efficient, and scalable web solutions.
	PO2	2	Moderately mapped since the primary focus is on technical utilization rather than deep problem formulation or extensive customer requirement analysis.
	PO3	2	Moderately mapped as While utilizing server-side technologies involves problem investigation and design, the direct application of innovative solutions incorporating ESG/SDG goals is generally not the primary focus of this CO.
	PO4	2	Moderately mapped as the students learn to select and apply appropriate tools and frameworks to build backend systems effectively.
	PO5	2	Moderately mapped since its main focus is on technical skill development rather than on leadership, communication, or formal use of Agile methodologies, the mapping with the PO emphasizing these soft skills and methodologies.
	PO8	1	Slightly mapped as the CO does not explicitly address change management skills or cultivate a structured approach to continuous learning and adapting to evolving technologies and work environments
C603.6	PO1	3	Substantially mapped as the Demonstrating the use of web services inherently involves applying core programming skills, logical reasoning, and architectural principles to design robust, efficient, and scalable software solutions.
	PO2	2	Moderately mapped as Web services development requires understanding needs and formulating solutions, but the depth of critical thinking and problem formulation.

	PO3	2	Moderately mapped as the students demonstrating use of web services can apply these technologies to innovative projects aligned with ESG/SDG goals.
	PO4	2	Moderately mapped as demonstrating the use of web services involves selecting and applying modern computational tools and some algorithmic logic.
	PO5	1	Slightly mapped as the students might work on web services in teams, offering limited opportunities to practice communication or leadership skills.
	PO8	2	Moderately mapped as the demonstrating use of web services involves engaging with evolving technologies and requires students to learn continuously.

Unit Wise Syllabus

Course Contents

Unit I	**Scripting Language - HTML**	**7 Hours**
HTML: Introduction to Internet, WWW, websites, HTML5 - HTML5 structure, headings, linking, images, lists, tables, forms, HTML5 - new elements, datalist element. **CSS:** Introduction, basic properties - text, list, border, font, Selectors - universal, element, id, class. CSS types -Inline, Internal and External style sheets.		
Unit II	**Client-Site Technology - JavaScript**	**7 Hours**
JavaScript: Using JS in an HTML (embedded, external), variables / data types, control structures, functions and dialog boxes, page redirect, cookies, events. JS objects -JavaScript object properties, methods, JavaScript properties and methods.		
Unit III	**Bootstrap & WordPress**	**7 Hours**
Bootstrap: Introduction to Bootstrap, Bootstrap basics, Bootstrap grids, Bootstrap themes, Bootstrap CSS, Bootstrap JS. **WordPress:** Installation, pages vs. posts, comments, sidebars and widgets, building menus, WordPress site.		
Unit IV	**Angular Js & jQuery**	**7 Hours**
Angular JS: Introduction to AngularJS, AngularJS expressions, AngularJS modules, AngularJS data binding, AngularJS scopes, AngularJS directives & events, AngularJS controllers, AngularJS filters, AngularJS services, AngularJS HTTP, AngularJS tables, AngularJS select, fetching data from MySQL, AngularJS validation, AngularJS API. **jQuery :** Introduction to jQuery, jQuery syntax, jQuery selectors, jQuery events, jQuery effects, jQuery HTML, jQuery traversing.		
Unit V	**Server Side Technology - PHP**	**7 Hours**
PHP: Introduction to PHP, using HTML forms, PHP form handling, get data sent from form fields through GET and POST method, form validation, sessions and cookies. **MySQL:** PHP-MySQL connection overview, 3 different approaches – procedure, object oriented, PDO,PHP-MySQL function to connect to database, access database, fetch result.		
Unit VI	**Web Services**	**7 Hours**
Web Services: Architecture of Web service, supporting standards for web services, SOAP (Simple Object Access Protocol), WSDL (Web Service Description Language), JAX -RPC (Java API for XML based Remote Procedure Call), UDDI (Universal Description, Discovery and Integration), comparing Web service to other architectures for remote processing.		
	Total Hours	**42**
Learning Resources		

Text Books:

1. Ivan Bayross, "HTML, DHTML, JavaScript, Perl & CGI", BPB Pub, 3rd Ed, ISBN:8176562742.
2. Shyam Seshadri, "Angular: Up and Running", O'REILLY Publication, Edition: 1st ISBN-101491999837.

Reference Books:

1. Thomas A. Powell, "The Complete Reference HTML", Mc. Graw Hill ISBN: 978-0-07-174170-5.

2. Danny Goodman, Michael Morrison, Paul Novitski, Tia Gustaff Rayl , "JavaScript Bible", 7th Edition. ISBN: 978-0-470-95280-1.
3. Achyut Godbole, Atul Kahate , "Web Technology" ,Tata Mcgraw Hill –ISBN 0- 07-0472 98-X .

MOOC Courses (Web Links):
1. Web Based Technologies and Multimedia Applications by Prof. P. V. Suresh, IGNOU: https://onlinecourses.swayam2.ac.in/nou20_cs05/preview

Internal Assessment Tools (40% Weightage in Course Outcome Attainment)

Sr. No.	Assessment Tool	Marks
1	Subjective Test	20
2	Problem Solving	20
3	Assignments	20
	Total	60 and converted to out of 40

External Assessment Tools (80% Weightage in Course Outcome Attainment)

Sr. No.	Assessment Tool	Marks
1	SEE	60
Total		**60**

EE: External Examination

Curriculum Booklet - Theory Course

Class: SY

Name of the Course – Web Programming

2024 Pattern

(With effect from 2024-2025)

Course in charges Module Coordinator HOD Mrs. Swati D. Ghule / Mr. Y. L. Puranik,

Dr. Prakash M. Kene, Dr. Prakash M. Kene

Teaching Plan

Sr. No.	Unit	Topics to be covered	Book Referred	Total Lecture Planned
1	I	**Scripting Language - HTML**	1	7
2	II	**Client-Site Technology - JavaScript**	1	7
3	III	**Bootstrap & WordPress**	1	7

4	IV	**Angular Js & jQuery**	2	7
5	V	**Server Side Technology - PHP**	1	7
6	VI	**Web Services**	1	7

Text Books:

1. Ivan Bayross, "HTML, DHTML, JavaScript, Perl & CGI", BPB Pub, 3rd Ed, ISBN:8176562742.

Reference Books:

1. Thomas A. Powell, "The Complete Reference HTML", Mc. Graw Hill ISBN: 978-0-07-174170-5.

Reference Web Links/ Research Paper/ Referred Book other than Mention in Syllabus:

https://onlinecourses.swayam2.ac.in/nou20_cs05/preview

Unit No.-I- Scripting Language - HTML

Lecture No.	Details of the Topic to be covered	References
1	Discuss on Syllabus, Introduction to Internet, WWW, websites	1
2	HTML5 - HTML5 structure, headings	1
3	linking, images, lists, tables, forms	1
4	HTML5 - new elements, datalist element.	1
5	Introduction to CSS, basic properties - text, list, border, font	1
6	Selectors - universal, element, id, class	1
7	CSS types -Inline, Internal and External style sheets.	1

Question Bank: Theory & Numerical Mapped to Course Outcome:

Q. 1	List new tags in HTML5 (any 5 tags with short description)
Q. 2	Explain table tag with example

Q. 3	What are the different types of Lists in HTML?
Q. 4	Differentiate between HTML and HTML5
Q. 5	What is form? Explain form components with example.
Q. 6	Explain uses of Classes in CSS with suitable example.
Q. 7	Differentiate between Inline and External Style Sheet.
Q. 8	What are different parts of CSS?
Q. 9	What is the difference between class and id selector?
Q. 10	Briefly explain CSS Syntax and Style.

Unit No. II - Client-Site Technology - JavaScript

Lecture No.	Details of the Topic to be covered	References
1	Using JS in an HTML (embedded, external),	1
2	Variables / data types, control structures	1
3	Functions and dialog boxes	1
4	Page redirect, cookies	1
5	Events. JS objects	1
6	JavaScript object properties	1
7	JavaScript properties and methods	1

Question Bank: Theory & Numerical
Mapped to Course Outcome:

Q. 1	Explain Event Handling in Javascript with suitable example.
Q. 2	Identify function object in JavaScript? Also how to create function objects that uses **Function** Constructor and **new** Operator.
Q. 3	How to use Array in JavaScript? Also explain types of Array?
Q. 4	Explain math and Date object in JavaScript
Q. 5	Differentiate between server side scripting and client side scripting.
Q. 6	What do you understand by use of JavaScript for Form validation?
Q. 7	Differentiate between functions and events with the help of example?

Q. 8	Briefly explain the various types of dialog boxes in Javascript with example.
Q. 9	State and explain session cookie.
Q. 10	Describe following events with their attribute and tags in Java Script a. Focus b. Click c. Load d. Submit

Unit No.-III- Bootstrap & WordPress

Lecture No.	Details of the Topic to be covered	References
1	Introduction to Bootstrap.	1
2	Bootstrap basics, Bootstrap grids	1
3	Bootstrap themes, Bootstrap CSS, Bootstrap JS.	1
4	Installation, pages vs. posts	1
5	Comments, sidebars and widgets	1
6	Building menus.	1
7	WordPress site.	1

Question Bank: Theory & Numerical
Mapped to Course Outcome:

Q. 1	**What are the key components of Bootstrap?**
Q. 2	What is a Bootstrap Container, and how does it work?
Q. 3	What do you know about the Bootstrap Grid System?
Q. 4	What is the difference between container and container-fluid classes in bootstrap?
Q. 5	What are Bootstrap's responsive utilities and how do you use them?
Q. 6	What are the main features of WordPress?
Q. 7	What is the main difference between WordPress posts and pages?
Q. 8	How do you handle database management in WordPress?

Q. 9	Can you explain how to use the WordPress REST API?
Q. 10	How to add a new menu in WordPress?

Unit No.-IV- Angular Js & jQuery

Lecture No.	Details of the Topic to be covered	References
1	Introduction to AngularJS, AngularJS expressions, AngularJS modules,	2
2	AngularJS data binding, AngularJS scopes, AngularJS directives & events	2
3	AngularJS controllers, AngularJS filters, AngularJS services, AngularJS HTTP, AngularJS tables,	2
4	AngularJS select, fetching data from MySQL, AngularJS validation, AngularJS API.	2
5	Introduction to jQuery, jQuery syntax	2
6	jQuery selectors, jQuery events, jQuery effects	2
7	jQuery HTML, jQuery traversing.	2

Question Bank: Theory & Numerical Mapped to Course Outcome:

Q. 1	Define MVC architecture with respect to AngularJS.
Q. 2	Explain AngularJS Controllers with example.
Q. 3	Explain AngularJS Modules with example.
Q. 4	What are directives? Name some of the most commonly used directives in AngularJS application
Q. 5	What is data binding in AngularJS?
Q. 6	Explain the concept of JQuery with selecting, creating, appending elements.
Q. 7	What are the selectors in jQuery? How many types of selectors in jQuery?
Q. 8	What is event delegation in jQuery?
Q. 9	What are the jQuery effects?

Q. 10	What are the advantages of jQuery?

Unit No.-V- Server Side Technology - PHP

Lecture No.	Details of the Topic to be covered	References
1	**Introduction to PHP, using HTML forms**.	1
2	**PHP form handling, get data sent from form fields through GET and POST method,**	1
3	Form validation, sessions and cookies	1
4	PHP-MySQL connection overview,	1
5	3 different approaches – procedure, object oriented,	1
6	PDO,PHP-MySQL function to connect to database	1
7	Access database, fetch result.	1

Question Bank: Theory & Numerical
Mapped to Course Outcome:

Q. 1	What is the use of header () function in PHP?
Q. 2	What is the difference between require and include?
Q. 3	What is session? How we can register variables into session?
Q. 4	Write short note on Cookie handling
Q. 5	Explain database connectivity in PHP with reference to MYSQL
Q. 6	How to create a database in MYSQL and how to create a table in it? Also explain How to insert data into it.
Q. 7	Explain of the following PHP/MySQL function (i) mysql_query() (ii) mysql_error (iii) isset()

Q. 8	Can you explain the difference between MyISAM and InnoDB storage engines?
Q. 9	What does the JOIN statement do in MySQL? Explain the different types of joins.
Q. 10	What is the difference between mysql_fetch_array and mysql_fetch_object?

Unit No.-VI- Web Services

Lecture No.	Details of the Topic to be covered	References
1	**Architecture of Web service,,**	1
2	Supporting standards for web services,	1
3	SOAP (Simple Object Access Protocol),	1
4	WSDL (Web Service Description Language),	1
5	JAX -RPC (Java API for XML based Remote Procedure Call),	1
6	UDDI (Universal Description, Discovery and Integration),	1
7	Comparing Web service to other architectures for remote processing	1

Question Bank: Theory & Numerical Mapped to Course Outcome:

Q. 1	Discuss the role of SOAP web service in details
Q. 2	What are Web services? Explain different types of Web Services.
Q. 3	Differentiate between SOAP and REST.
Q. 4	What is semantic web service? Name any four web services technologies
Q. 5	What is WSDL? Highlight its significance.
Q. 6	What is RPC? List out the component of web service architecture.
Q. 7	What is semantic web service? 7. Name any four web services technologies
Q. 8	What is JAX-RPC? Explain in detail with the help of diagram

Q. 9	Explain the UDDI Registries? What are the uses of UDDI Registry?
Q. 10	Describe the WSDL Structure and its life cycle in detail?